

GRADE 4 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 5) • Forward 2 (Grade 6)
64 question sets • 1600 questions

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Interpret a multiplication equation as a comparison

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Compare the size of a product to the size of one factor	5.NF.B.5.a
FORWARD 2	Use ratio language to describe a relationship	6.RP.A.1

SET 1 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Which statement matches $24 = 6 \times 4$?

- A.** 24 is 6 more than 4.
- B.** 6 is 4 times as many as 24.
- C.** 24 is 4 more than 6.
- D.** 24 is 6 times as many as 4.

Write a comparison statement for $45 = 9 \times 5$.

Statement: _____

True or false?

$56 = 7 \times 8$ can mean that 56 is 8 times as many as 7.

A. True

B. False

True or false?

$18 = 3 \times 6$ means that 18 is 3 more than 6.

A. True

B. False

Which equation means "63 is 7 times as many as 9"?

A. $63 = 7 + 9$

B. $7 = 63 \times 9$

C. $63 = 7 \times 9$

D. $63 = 9 - 7$

SET 1 • 4.OA.A.1

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

What is the total in 4 groups of 6?

Total: _____

True or false?

5×3 can mean 5 groups with 3 objects in each group.

A. True

B. False

Which expression shows 3 groups of 7?

A. $7 - 3$

B. $3 + 7$

C. $7 \div 3$

D. 3×7

Write a multiplication equation for 6 groups of 2.

Equation: _____

True or false?

2×8 means 2 more than 8.

A. True

B. False

SET 1 • 4.OA.A.1

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 5$$

Product: _____

True or false?

$$7 \times 8 = 54$$

A. True

B. False

$$6 \times 4 = ?$$

A. 10

B. 20

C. 24

D. 28

Multiply.

$$8 \times 6$$

Product: _____

Multiply.

$$7 \times 9$$

Product: _____

SET 1 • 4.OA.A.1

FORWARD 1

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, how does 3×25 compare with 25?

- A.** It is 3 more.
- B.** It is smaller than 25.
- C.** It is equal to 25.
- D.** It is 3 times as large.

True or false?

$\frac{1}{2} \times 40$ is less than 40.

A. True

B. False

A rope is 12 feet long. A second rope is 4 times as long.
How long is the second rope?

Length: _____

True or false?

5×18 is less than 18.

A. True

B. False

Which product is greater than 30?

A. $\frac{1}{2} \times 30$

B. 2×30

C. $\frac{3}{4} \times 30$

D. $\frac{1}{3} \times 30$

SET 1 • 4.OA.A.1

FORWARD 2

Use ratio language to describe a relationship • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A class has 12 girls and 4 boys.

Write the ratio of girls to boys.

Ratio: _____

True or false?

If there are 3 times as many cats as dogs, the ratio of cats to dogs is 3 : 1.

A. True

B. False

A bag has 5 red marbles for every 2 blue marbles.

Which statement describes the ratio?

- A.** There are 5 more red marbles than blue.
- B.** For every 5 red marbles there are 2 blue marbles.
- C.** There are 2 times as many red marbles as blue.
- D.** There are 7 red marbles.

There are 18 apples and 6 pears.

How many apples are there for every 1 pear?

Apples: _____

True or false?

The ratio 2 : 8 means there are 2 times as many of the first quantity as the second.

- A. True
- B. False

Write a comparison statement as a multiplication equation

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Multiplication word problems within 100	3.OA.A.3
BACKWARD 2	Find the unknown number in a multiplication equation	3.OA.A.4
FORWARD 1	Write numerical expressions from words	5.OA.A.2
FORWARD 2	Write expressions that use a variable	6.EE.A.2.a

SET 2 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Write an equation.

32 is 4 times as many as 8.

Equation: _____

Kai has 5 stickers. Lena has 3 times as many stickers as Kai.

Which equation gives Lena's stickers, s ?

A. $s = 3 \times 5$

B. $s = 5 - 3$

C. $s = 3 + 5$

D. $s = 5 \div 3$

Write an equation.

54 is 6 times as many as 9.

Equation: _____

True or false?

"40 is 8 times as many as 5" can be written as
 $40 = 8 + 5$.

A. True

B. False

A tree is 7 times as tall as a 2-meter bush.
Write an equation for the tree's height, t .

Equation: _____

SET 2 • 4.OA.A.1

BACKWARD 1

Multiplication word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 boxes with 8 crayons in each box.
How many crayons are there?

Crayons: _____

True or false?

5 bags with 9 apples in each bag hold 45 apples.

A. True

B. False

A garden has 4 rows with 7 plants in each row.

How many plants are there?

- A.** 21
- B.** 32
- C.** 28
- D.** 11

Each pack has 3 juice boxes.

How many juice boxes are in 9 packs?

Juice boxes: _____

True or false?

8 teams with 5 players each have 13 players in all.

A. True

B. False

SET 2 • 4.OA.A.1

BACKWARD 2

Find the unknown number in a multiplication equation • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 6 = 30$$

Unknown: _____

True or false?

If $4 \times n = 36$, then $n = 8$.

A. True

B. False

$$8 \times ? = 56$$

A. 8

B. 7

C. 6

D. 48

Find the unknown number.

$$3 \times ? = 27$$

Unknown: _____

True or false?

If $m \times 7 = 42$, then $m = 6$.

A. True

B. False

SET 2 • 4.OA.A.1

FORWARD 1

Write numerical expressions from words • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression.

3 times the sum of 8 and 4

Expression: _____

Which expression means "add 9 and 6, then multiply by 2"?

A. $9 \times 6 + 2$

B. $9 + 6 \times 2$

C. $(9 + 6) \times 2$

D. $2 + 9 \times 6$

True or false?

$5 \times (12 - 3)$ is 5 times as large as $12 - 3$.

A. True

B. False

Write an expression.

4 times as large as $25 + 15$

Expression: _____

True or false?

$2 \times (7 + 1)$ is 2 more than $7 + 1$.

A. True

B. False

SET 2 • 4.OA.A.1

FORWARD 2

Write expressions that use a variable • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for 5 times a number n .

Expression: _____

Which expression means "8 more than y "?

A. $y + 8$

B. $8 - y$

C. $y - 8$

D. $8y$

Maya has m marbles. Jo has 4 times as many marbles as Maya.

Write an expression for Jo's marbles.

Expression: _____

True or false?

"3 times the sum of x and 2" can be written as $3(x + 2)$.

A. True

B. False

True or false?

"7 less than k" can be written as $7 - k$.

A. True

B. False

Solve multiplicative comparison word problems by multiplying

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Multiplication and division word problems within 100	3.OA.A.3
BACKWARD 2	Multiply in either order	3.OA.B.5
FORWARD 1	Solve real-world problems by multiplying by a fraction or mixed nu...	5.NF.B.6
FORWARD 2	Solve unit rate problems	6.RP.A.3.b

SET 3 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A puppy weighs 6 pounds. Its mother weighs 8 times as much.

How much does the mother weigh?

Puppy

6

Mother

6	6	6	6	6	6	6	6
---	---	---	---	---	---	---	---

Weight: _____

Sam read 9 pages. Ana read 5 times as many pages as Sam.

How many pages did Ana read?

Pages: _____

A red ribbon is 7 inches long. A blue ribbon is 6 times as long.

How long is the blue ribbon?

A. 13 inches

B. 48 inches

C. 1 inch

D. 42 inches

A small tank holds 9 gallons. A large tank holds 7 times as much.

How much does the large tank hold?

Gallons: _____

True or false?

A bike costs \$80. A used car costs 100 times as much. The car costs \$800.

A. True

B. False

SET 3 • 4.OA.A.2

BACKWARD 1

Multiplication and division word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

7 vans each carry 8 students.

How many students are there in all?

Students: _____

36 crayons are shared equally by 4 children.

How many crayons does each child get?

Crayons: _____

True or false?

6 rows of 6 chairs make 36 chairs.

A. True

B. False

A book has 9 chapters with 9 pages in each chapter.

How many pages does it have?

A. 18

B. 72

C. 81

D. 90

True or false?

40 cards in 5 equal piles makes 9 cards in each pile.

A. True

B. False

SET 3 • 4.OA.A.2

BACKWARD 2

Multiply in either order • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times 8 = 8 \times 6$$

A. True

B. False

Fill in the blank.

$$9 \times 4 = 4 \times \underline{\quad}$$

Blank: _____

Which is equal to 7×5 ?

A. 5×7

B. $7 + 5$

C. 75

D. $5 + 5 + 5$

$$3 \times 9 = 27.$$

What is 9×3 ?

Product: _____

True or false?

$$8 \times 2 = 2 + 8$$

A. True

B. False

SET 3 • 4.OA.A.2

FORWARD 1

Solve real-world problems by multiplying by a fraction or mixed number •
Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A pup weighs 6 pounds. An adult dog weighs $4\frac{1}{2}$ times as much.

How much does the adult weigh?

Weight: _____

A path is 12 miles long. Kim walked $\frac{2}{3}$ of the path.

How far did she walk?

Distance: _____

A red rope is 8 feet long. A blue rope is $1\frac{1}{4}$ times as long.

How long is the blue rope?

A. 10 feet

B. $9\frac{1}{4}$ feet

C. 2 feet

D. 6 feet

True or false?

A 10-cm plant grows until it is $2\frac{1}{2}$ times as tall. It is then 25 cm tall.

- A. True
- B. False

A recipe uses 3 cups of flour. Mia makes $\frac{1}{2}$ of the recipe.

How much flour does she use?

Flour: _____

SET 3 • 4.OA.A.2

FORWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

5 notebooks cost \$15.

At the same price, how much do 8 notebooks cost?

Cost: _____

True or false?

If 4 pounds of grapes cost \$10, then 1 pound costs \$2.50.

A. True

B. False

A car goes 180 miles in 3 hours at a steady speed.

How far does it go in 5 hours?

- A.** 240 miles
- B.** 360 miles
- C.** 300 miles
- D.** 185 miles

A printer prints 42 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

True or false?

If 3 tickets cost \$27, then 7 tickets cost \$56.

- A. True
- B. False

Solve multiplicative comparison word problems by dividing

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find the unknown number in a multiplication or division equation	3.OA.A.4
FORWARD 1	Solve division word problems with two-digit divisors	5.NBT.B.6
FORWARD 2	Solve one-step equations of the form $px = q$	6.EE.B.7

SET 4 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A dog weighs 48 pounds. That is 6 times as much as a cat weighs.

How much does the cat weigh?

Weight: _____

Leo has 72 cards. That is 9 times as many cards as Ty has.

How many cards does Ty have?

Cards: _____

A building is 60 feet tall. That is 5 times as tall as a tree.
How tall is the tree?

- A.** 300 feet
- B.** 55 feet
- C.** 65 feet
- D.** 12 feet

A jacket costs \$56. A hat costs \$8.

The jacket costs how many times as much as the hat?

Times as much: _____

True or false?

Mo ran 36 laps. That is 4 times as many laps as Ed ran. Ed ran 9 laps.

- A. True
- B. False

SET 4 • 4.OA.A.2

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $42 \div 6$ by thinking $6 \times ? = 42$.

Quotient: _____

True or false?

$54 \div 9 = 6$ because $9 \times 6 = 54$.

A. True

B. False

Which fact helps you find $63 \div 7$?

A. $9 + 7 = 16$

B. $63 - 7 = 56$

C. $7 \times 9 = 63$

D. $7 \times 7 = 49$

Divide.

$$32 \div 8$$

Quotient: _____

True or false?

$48 \div 6 = 9$ because $6 \times 9 = 48$.

A. True

B. False

SET 4 • 4.OA.A.2

BACKWARD 2

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$5 \times ? = 40$$

Unknown: _____

True or false?

If $27 \div ? = 3$, the unknown number is 9.

A. True

B. False

$$? \times 9 = 72$$

A. 63

B. 9

C. 81

D. 8

Find the unknown number.

$$? \div 4 = 6$$

Unknown: _____

True or false?

If $6 \times ? = 36$, the unknown number is 5.

A. True

B. False

SET 4 • 4.OA.A.2

FORWARD 1

Solve division word problems with two-digit divisors • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

A truck carries 384 boxes. That is 12 times as many boxes as a van carries.

How many boxes does the van carry?

Boxes: _____

A school has 864 students in 24 classes of the same size.

How many students are in each class?

A. 38

B. 36

C. 34

D. 840

True or false?

$$1,050 \div 15 = 70$$

A. True

B. False

Divide.

$$2,016 \div 36$$

Quotient: _____

True or false?

A farm has 630 trees in rows of 18 trees. There are 45 rows.

A. True

B. False

SET 4 • 4.OA.A.2

FORWARD 2

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for c.

$$6c = 48$$

$$c = \underline{\hspace{10cm}}$$

True or false?

The solution of $9x = 72$ is $x = 8$.

A. True

B. False

Solve for n.

$$5n = 65$$

A. 325

B. 60

C. 70

D. 13

A dog weighs 48 pounds, which is 6 times the weight c of a cat.

Write and solve an equation for c .

Equation: _____

$c =$ _____

True or false?

The solution of $4y = 36$ is $y = 32$.

A. True

B. False

Tell multiplicative comparison from additive comparison

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Compare quantities by adding or subtracting	2.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Interpret numerical expressions without evaluating them	5.OA.A.2
FORWARD 2	Tell an additive relationship from a multiplicative one in tables a...	6.EE.C.9

SET 5 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

Ana has 4 shells. Ben has 12 shells.

Which statement is true?

- A.** Ben has 3 times as many shells as Ana.
- B.** Ben has 3 more shells than Ana.
- C.** Ana has 8 times as many shells as Ben.
- D.** Ben has 4 more shells than Ana.

The red team scored 24 points. The blue team scored 6 points.

How many more points did the red team score? How many times as many points?

How many more: _____

How many times as many: _____

True or false?

"8 more than 5" is 40.

A. True

B. False

Which question is answered by multiplying?

A. A cat is 3 years old. A dog is 4 times as old. How old is the dog?

B. A cat ate 4 treats and then 3 more. How many did it eat?

C. A cat is 3 years old. A dog is 4 years older. How old is the dog?

D. A cat is 7 years old. A dog is 3 years old. How much older is the cat?

True or false?

A pencil is 6 inches long. A ruler that is 2 times as long is 12 inches, but a crayon that is 2 inches longer than the pencil is 8 inches.

- A.** True
- B.** False

SET 5 • 4.OA.A.2

BACKWARD 1

Compare quantities by adding or subtracting • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Jo has 15 stamps. Ty has 9 stamps.
How many more stamps does Jo have?

Stamps: _____

True or false?

Mia has 8 beads. That is 5 more beads than Kim has. Kim has 3 beads.

A. True

B. False

A red rope is 34 cm long. A green rope is 18 cm long.
How much longer is the red rope?

- A.** 14 cm
- B.** 16 cm
- C.** 52 cm
- D.** 26 cm

A tower has 27 blocks. Another tower has 8 more blocks.

How many blocks does it have?

Blocks: _____

True or false?

Sam has 20 cards. Lee has 6 fewer cards. Lee has 26 cards.

A. True

B. False

SET 5 • 4.OA.A.2

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$4 \times 9$$

Product: _____

True or false?

$$6 \times 7 = 42$$

A. True

B. False

$$3 \times 8 = ?$$

A. 21

B. 24

C. 11

D. 32

Multiply.

$$5 \times 6$$

Product: _____

True or false?

$$8 \times 8 = 16$$

A. True

B. False

SET 5 • 4.OA.A.2

FORWARD 1

Interpret numerical expressions without evaluating them • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$6 \times (25 + 9)$ is 6 times as large as $25 + 9$.

A. True

B. False

How does $4 \times (17 - 3)$ compare with $17 - 3$?

- A.** It is 4 times as large.
- B.** It is the same.
- C.** It is 4 less.
- D.** It is 4 more.

True or false?

$(48 + 6) + 3$ is 3 times as large as $48 + 6$.

A. True

B. False

Without calculating, how does $10 \times (325 + 19)$ compare with $325 + 19$?

Answer: _____

Which expression is 2 more than 9×7 ?

A. $9 \times (7 + 2)$

B. $2 \times 9 \times 7$

C. $9 \times 7 + 2$

D. $(9 + 2) \times 7$

SET 5 • 4.OA.A.2

FORWARD 2

Tell an additive relationship from a multiplicative one in tables and equations •
Grade 6 • Expressions & Equations • 6.EE.C.9

Which equation shows that y is always 4 times x ?

A. $y = 4x$

B. $y = x + 4$

C. $y = x - 4$

D. $x = 4y$

True or false?

The table fits the equation $y = 5x$.

x	1	2	3
y	5	6	7

A. True

B. False

Write an equation for y in terms of x .

x	2	4	6
y	6	12	18

Equation: _____

Write an equation for y in terms of x .

x	2	4	6
y	5	7	9

Equation: _____

True or false?

In $y = x + 6$, y is always 6 more than x .

A. True

B. False

Solve multistep word problems and write an equation with a letter for the unknown

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Two-step word problems	3.OA.D.8
BACKWARD 2	Add and subtract within 1000	3.NBT.A.2
FORWARD 1	Evaluate expressions with parentheses, brackets, and braces	5.OA.A.1
FORWARD 2	Use variables to write expressions for real-world problems	6.EE.B.6

SET 6 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A school bought 8 packs of 24 pencils. It gave away 57 pencils.

Write an equation with a letter for the number of pencils left. Then solve it.

Equation: _____

Pencils left: _____

Ty has \$250. He buys 3 games that cost \$45 each.

Write an equation with a letter for the money left. Then solve it.

Equation: _____

Money left: _____

A farmer has 6 crates with 35 apples in each crate. She sells 98 apples.

Which equation gives the number of apples left, a ?

A. $a = 6 \times 35 - 98$

B. $a = 6 + 35 - 98$

C. $a = 98 - 6 \times 35$

D. $a = 6 \times (35 - 98)$

Tickets cost \$9 for adults and \$6 for children. A group buys 4 adult tickets and 7 child tickets.

Write an equation with a letter for the total cost. Then solve it.

Equation: _____

Total cost: _____

True or false?

A baker makes 128 rolls and puts them in bags of 8. She sells 11 bags. The equation $b = 128 \div 8 - 11$ gives the bags left, and $b = 5$.

A. True

B. False

SET 6 • 4.OA.A.3

BACKWARD 1

Two-step word problems • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Ann had 45 beads. She bought 3 packs with 8 beads in each pack.

How many beads does she have now?

Beads: _____

True or false?

Jo has 50 cents. She spends 4 groups of 5 cents.
She has 30 cents left.

- A.** True
- B.** False

There are 7 rows of 6 chairs. 15 chairs are taken.
How many chairs are empty?

- A.** 13
- B.** 42
- C.** 57
- D.** 27

Leo has 3 boxes of 9 markers. He gives away 8 markers.
How many markers does he have left?

Markers: _____

True or false?

6 bags of 9 oranges, with 10 oranges eaten, leaves 50 oranges.

A. True

B. False

SET 6 • 4.OA.A.3

BACKWARD 2

Add and subtract within 1000 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.2

Add.

$$468 + 257$$

Sum: _____

Subtract.

$$803 - 469$$

Difference: _____

True or false?

$$350 + 475 = 825$$

A. True

B. False

$$612 - 238 = ?$$

A. 484

B. 374

C. 386

D. 850

True or false?

$$900 - 345 = 665$$

A. True

B. False

SET 6 • 4.OA.A.3

FORWARD 1

Evaluate expressions with parentheses, brackets, and braces • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$8 \times 24 - 57$$

Value: _____

Evaluate.

$$250 - 3 \times 45$$

Value: _____

Evaluate.

$$(36 + 12) \div 6$$

A. 38

B. 8

C. 12

D. 42

True or false?

$$4 \times (9 + 6) = 42$$

A. True

B. False

Evaluate.

$$[40 - (5 \times 6)] \times 3$$

Value: _____

SET 6 • 4.OA.A.3

FORWARD 2

Use variables to write expressions for real-world problems • Grade 6 •
Expressions & Equations • 6.EE.B.6

Pens cost \$3 each.

Write an expression for the cost of p pens.

Expression: _____

Ty has 40 stickers and gives away s stickers.
Which expression shows how many he has left?

A. $40 - s$

B. $s - 40$

C. $40s$

D. $40 + s$

A movie ticket costs \$9 and a popcorn costs \$4.

Write an expression for the cost of t tickets and one popcorn.

Expression: _____

True or false?

If n is the number of packs with 12 cards in each pack, then $12n$ is the total number of cards.

A. True

B. False

True or false?

The expression $5 + b$ means 5 times b .

A. True

B. False

Interpret remainders in word problems

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Quotients as shares or as groups	3.OA.A.2
BACKWARD 2	Division facts within 100	3.OA.C.7
FORWARD 1	Interpret a fraction as division to write a remainder as a fraction	5.NF.B.3
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 7 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

29 students ride in cars that hold 4 students each.
How many cars are needed?

Cars: _____

A baker has 50 muffins. She puts 6 muffins in each box.
How many boxes can she fill completely?

Boxes: _____

Ella shares 35 stickers equally among 4 friends.

How many stickers are left over?

A. 8

B. 3

C. 4

D. 1

A rope 46 feet long is cut into 5-foot pieces.

How many full pieces are there, and how much rope is left over?

Full pieces: _____

Left over: _____

True or false?

70 people need tables that seat 8 people each.
They need 8 tables.

A. True

B. False

SET 7 • 4.OA.A.3

BACKWARD 1

Quotients as shares or as groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

24 pencils are shared equally by 6 students.

How many pencils does each student get?

Pencils: _____

True or false?

$30 \div 5$ can mean 30 objects put into groups of 5.

A. True

B. False

Which question does $18 \div 3$ answer?

- A.** What is 3 more than 18?
- B.** How many groups of 3 are in 18?
- C.** What is 18 take away 3?
- D.** How many are in 18 groups of 3?

40 crayons are put into boxes of 8.

How many boxes are filled?

Boxes: _____

True or false?

$12 \div 4$ means 12 groups of 4.

A. True

B. False

SET 7 • 4.OA.A.3

BACKWARD 2

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$45 \div 5 = 9$$

A. True

B. False

$$64 \div 8 = ?$$

A. 56

B. 7

C. 9

D. 8

Divide.

$$36 \div 4$$

Quotient: _____

True or false?

$$81 \div 9 = 8$$

A. True

B. False

SET 7 • 4.OA.A.3

FORWARD 1

Interpret a fraction as division to write a remainder as a fraction • Grade 5 •
Number & Operations–Fractions • 5.NF.B.3

3 friends share 7 pizzas equally.

How much pizza does each friend get? Write a mixed number.

Pizza: _____

True or false?

When 9 feet of ribbon is cut into 4 equal pieces, each piece is $2\frac{1}{4}$ feet long.

- A. True
- B. False

5 sandwiches are shared equally by 2 people.

How much does each person get?

- A.** $\frac{2}{5}$ sandwich
- B.** 2 sandwiches
- C.** $2\frac{1}{2}$ sandwiches
- D.** 3 sandwiches

Write $17 \div 5$ as a mixed number.

Mixed number: _____

True or false?

$$11 \div 4 = 2\frac{1}{4}$$

A. True

B. False

SET 7 • 4.OA.A.3

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$4,368 \div 52$$

Quotient: _____

Divide.

$$7,695 \div 81$$

Quotient: _____

Divide.

$$9,072 \div 36$$

A. 25

B. 252

C. 242

D. 262

Divide. Write the quotient and the remainder.

$$1,000 \div 23$$

Quotient: _____

Remainder: _____

True or false?

$$3,240 \div 45 = 72$$

A. True

B. False

Use rounding and mental math to check whether an answer is reasonable

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Round whole numbers to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Check answers with rounding	3.OA.D.8
FORWARD 1	Round decimals to estimate	5.NBT.A.4
FORWARD 2	Use estimates to check computations with multi-digit decimals —...	6.NS.B.3

SET 8 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Round each number to the nearest hundred. Then estimate the sum.

$$512 + 289$$

Rounded numbers: _____

Estimate: _____

Mia says $48 \times 6 = 188$.

Round 48 to the nearest ten and estimate 48×6 . Is her answer reasonable?

Estimate: _____

Reasonable? _____

Round each number to the nearest ten. Which is the estimate of $73 + 28 + 41$?

A. 150

B. 142

C. 140

D. 130

A store had 1,215 shirts and sold 487 of them.

Round each number to the nearest hundred to estimate how many shirts are left.

Estimate: _____

True or false?

Rounding each number to the nearest hundred,
 $395 + 612$ is about 900.

A. True

B. False

SET 8 • 4.OA.A.3

BACKWARD 1

Round whole numbers to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1

Round 67 to the nearest ten.

Rounded: _____

True or false?

348 rounded to the nearest hundred is 400.

A. True

B. False

Round 452 to the nearest ten.

A. 400

B. 460

C. 500

D. 450

Round 815 to the nearest hundred.

Rounded: _____

True or false?

75 rounded to the nearest ten is 80.

A. True

B. False

SET 8 • 4.OA.A.3

BACKWARD 2

Check answers with rounding • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

Rounding each number to the nearest ten,
 $38 + 51$ is about 90.

A. True

B. False

Leo says $297 + 102 = 499$. Rounding to the nearest hundred gives $300 + 100 = 400$.

What does the estimate show?

- A.** An estimate cannot check a sum.
- B.** His answer is too big.
- C.** His answer is correct.
- D.** His answer is too small.

Round each number to the nearest ten. Then estimate.

$$62 - 29$$

Estimate: _____

True or false?

Rounding each number to the nearest hundred,
 $710 - 290$ is about 600.

A. True

B. False

Round each number to the nearest ten. Then estimate.

$$47 + 33$$

Estimate: _____

SET 8 • 4.OA.A.3

FORWARD 1

Round decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round each number to the nearest whole number. Then estimate.

$$4.8 + 7.3$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $9.6 - 3.2$ is about 7.

- A. True
- B. False

Round 6.47 to the nearest tenth.

A. 6.4

B. 6

C. 6.5

D. 7

Round each number to the nearest whole number. Then estimate.

$$2.9 \times 4.1$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $12.5 + 6.5$ is about 18.

A. True

B. False

FORWARD 2

Use estimates to check computations with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Round each number to the nearest whole number. Use your estimate to decide whether $18.75 + 6.4 = 25.15$ is reasonable.

Estimate: _____

Reasonable? _____

True or false?

Rounding each number to the nearest whole number, 3.9×5.2 is about 20, so a product of 202.8 is not reasonable.

A. True

B. False

Ann computed $45.6 \div 3.8 = 1.2$. Rounding each number to the nearest whole number gives $46 \div 4$, which is about 11.

What does this show?

- | | | | |
|------------------------------------|--|--|----------------------------------|
| A. Her answer is too large. | B. The estimate cannot be used. | C. Her answer is far too small. | D. Her answer is correct. |
|------------------------------------|--|--|----------------------------------|

Multiply.

$$12.4 \times 3.5$$

Product: _____

True or false?

Rounding each number to the nearest whole number, 7.1×2.9 is about 21, so a product of 2.059 is reasonable.

- A.** True
- B.** False

Find all factor pairs of a whole number from 1 to 100

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Find an unknown factor	3.OA.B.6
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Use multiples to find a common denominator	5.NF.A.1
FORWARD 2	Find the greatest common factor	6.NS.B.4

SET 9 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 24.

Factor pairs: _____

List all the factor pairs of 45.

Factor pairs: _____

Which is NOT a factor pair of 60?

A. 7×9

B. 5×12

C. 6×10

D. 4×15

True or false?

36 has exactly 5 factor pairs.

A. True

B. False

How many factors does 18 have? List them.

How many: _____

Factors: _____

SET 9 • 4.OA.B.4

BACKWARD 1

Find an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find the unknown factor.

$$4 \times ? = 24$$

Factor: _____

True or false?

$3 \times 8 = 24$, so 3 and 8 are a factor pair of 24.

A. True

B. False

Which number times 5 equals 45?

- A.** 9
- B.** 50
- C.** 40
- D.** 8

Find the unknown factor.

$$? \times 7 = 28$$

Factor: _____

True or false?

$$6 \times 7 = 48$$

A. True

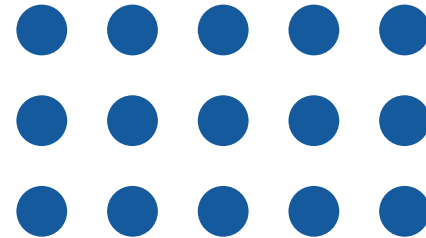
B. False

SET 9 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many dots are in the array?



Dots: _____

True or false?

An array with 4 rows of 5 dots has 20 dots.

- A. True
- B. False

Which addition equation matches an array with 2 rows of 4?

A. $4 - 2 = 2$

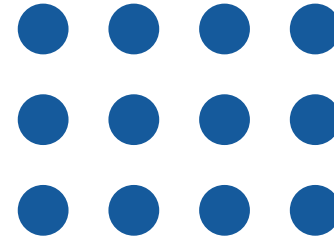
B. $2 + 2 = 4$

C. $4 + 4 = 8$

D. $2 + 4 = 6$

Write an addition equation for the array.

Equation: _____



True or false?

An array with 5 rows of 3 dots has 8 dots.

- A. True
- B. False

SET 9 • 4.OA.B.4

FORWARD 1

Use multiples to find a common denominator • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Find the least common denominator of $\frac{1}{4}$ and $\frac{1}{6}$.

Denominator: _____

True or false?

$\frac{2}{3}$ and $\frac{3}{4}$ can both be written with a denominator of 12.

- A. True
- B. False

Which is a common denominator of $\frac{1}{3}$ and $\frac{2}{5}$?

- A.** 3
- B.** 15
- C.** 10
- D.** 8

Add.

$$\frac{1}{2} + \frac{1}{3}$$

Sum: _____

True or false?

A common denominator of $\frac{1}{4}$ and $\frac{1}{10}$ is 14.

A. True

B. False

SET 9 • 4.OA.B.4

FORWARD 2

Find the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

Find the greatest common factor of 18 and 45.

GCF: _____

What is the greatest common factor of 16 and 40?

- A.** 2
- B.** 16
- C.** 4
- D.** 8

True or false?

The greatest common factor of 14 and 35 is 7.

- A. True
- B. False

True or false?

The greatest common factor of 20 and 30 is 5.

A. True

B. False

A whole number is a multiple of each of its factors; find multiples

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Skip count and find patterns in the multiplication table	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Generate two patterns and compare their terms	5.OA.B.3
FORWARD 2	Find the least common multiple	6.NS.B.4

SET 10 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first six positive multiples of 7.

Multiples: _____

True or false?

Because 6 is a factor of 42, 42 is a multiple of 6.

A. True

B. False

Which number is a multiple of 9?

A. 39

B. 19

C. 54

D. 92

Is 84 a multiple of 6? Explain.

Answer: _____

Explanation: _____

True or false?

70 is a multiple of 4.

A. True

B. False

SET 10 • 4.OA.B.4

BACKWARD 1

Skip count and find patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

Skip count by 6.

6, 12, 18, __, __

Next two: _____

True or false?

Every number in the 5s row of the multiplication table ends in 0 or 5.

- A.** True
- B.** False

Which number is in the 4s row of the multiplication table?

- A.** 18
- B.** 30
- C.** 28
- D.** 26

What comes next?

9, 18, 27, 36, ____

Next: _____

True or false?

Every number in the 3s row of the multiplication table is odd.

- A.** True
- B.** False

SET 10 • 4.OA.B.4

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

16 is even.

A. True

B. False

Is 23 odd or even?

Answer: _____

Which number is even?

A. 27

B. 38

C. 41

D. 15

Write 14 as the sum of two equal addends.

Sum: _____

True or false?

An even number of objects cannot be put into two equal groups.

- A.** True
- B.** False

SET 10 • 4.OA.B.4

FORWARD 1

Generate two patterns and compare their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12.

Write the first five terms of each pattern.

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12. Each term of B is 3 times the matching term of A.

A. True

B. False

Pattern A: 0, 5, 10, 15. Pattern B: 0, 10, 20, 30.

Which ordered pair (A, B) comes from the patterns?

A. (15, 30)

B. (15, 20)

C. (30, 15)

D. (10, 30)

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 10.

What is the 5th term of Pattern B?

Term: _____

True or false?

Each term of Pattern B is 4 times the matching term of Pattern A. When A is 9, B is 13.

A. True

B. False

SET 10 • 4.OA.B.4

FORWARD 2

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

Find the least common multiple of 5 and 7.

LCM: _____

What is the least common multiple of 4 and 10?

A. 20

B. 2

C. 14

D. 40

True or false?

The least common multiple of 9 and 12 is 36.

A. True

B. False

True or false?

The least common multiple of 3 and 6 is 18.

A. True

B. False

Decide whether a whole number from 1 to 100 is prime or composite

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Find the unknown number in a multiplication or division equation	3.OA.A.4
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Evaluate expressions with grouping symbols	5.OA.A.1
FORWARD 2	Write and evaluate expressions with whole-number exponents	6.EE.A.1

SET 11 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

Which number is prime?

A. 49

B. 33

C. 29

D. 27

True or false?

51 is a prime number.

A. True

B. False

Is 39 prime or composite? Explain.

Answer: _____

Explanation: _____

List all the prime numbers between 20 and 40.

Primes: _____

True or false?

A composite number has more than two factors.

A. True

B. False

SET 11 • 4.OA.B.4

BACKWARD 1

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 3 = 21$$

Unknown: _____

True or false?

If $7 \times ? = 49$, the unknown number is 7.

A. True

B. False

$$9 \times ? = 54$$

A. 7

B. 45

C. 6

D. 5

Find the unknown number.

$$35 \div ? = 5$$

Unknown: _____

True or false?

If $? \times 4 = 32$, the unknown number is 9.

A. True

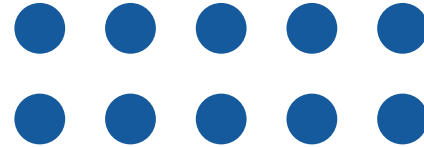
B. False

SET 11 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation for the array.



Equation: _____

True or false?

5 dots can make a rectangle only as 1 row of 5 or 5 rows of 1.

A. True

B. False

Which array has 12 dots?

- A.** 3 rows of 4
- B.** 4 rows of 4
- C.** 3 rows of 5
- D.** 2 rows of 5

How many dots are in an array with 4 rows of 4?

Dots: _____

True or false?

An array with 2 rows of 3 dots has 5 dots.

- A. True
- B. False

SET 11 • 4.OA.B.4

FORWARD 1

Evaluate expressions with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times 3 \times 7$$

Value: _____

Evaluate.

$$2 \times (5 \times 5)$$

Value: _____

Evaluate.

$$(2 \times 2) \times (3 \times 3)$$

A. 10

B. 36

C. 24

D. 12

True or false?

$$3 \times (2 \times 11) = 66$$

A. True

B. False

True or false?

$$[2 \times (3 + 5)] \times 2 = 22$$

A. True

B. False

SET 11 • 4.OA.B.4

FORWARD 2

Write and evaluate expressions with whole-number exponents • Grade 6 •
Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^3 \times 3$$

Value: _____

Write $2 \times 2 \times 2 \times 2 \times 5$ using an exponent.

Expression: _____

Which expression equals 72?

A. 6^2

B. $2^2 \times 3^3$

C. $2^3 \times 3^2$

D. $2 \times 3 \times 6$

True or false?

$$3^2 \times 5 = 45$$

A. True

B. False

True or false?

$$2^5 = 10$$

A. True

B. False

Generate a number or shape pattern that follows a given rule

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Arithmetic patterns	3.OA.D.9
BACKWARD 2	Add or subtract 10 or 100 mentally	2.NBT.B.8
FORWARD 1	Form ordered pairs from two patterns	5.OA.B.3
FORWARD 2	Write an equation for a pattern with dependent and independent v...	6.EE.C.9

SET 12 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 7." The pattern starts at 4.
Write the first five terms.

Terms: _____

The rule is "multiply by 3." The pattern starts at 2.
Write the first four terms.

Terms: _____

Which pattern follows the rule "subtract 6," starting at 50?

A. 50, 45, 40, 35

B. 50, 44, 38, 32

C. 50, 56, 62, 68

D. 44, 38, 32, 26

True or false?

The rule "add 9," starting at 15, gives 15, 24, 33, 43.

A. True

B. False

In a shape pattern, each figure has 2 more squares than the one before. The 1st figure has 3 squares.

How many squares are in the 6th figure?

Squares: _____

SET 12 • 4.OA.C.5

BACKWARD 1

Arithmetic patterns • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

What comes next?

12, 16, 20, 24, ____

Next: _____

True or false?

The pattern 5, 10, 15, 20 adds 5 each time.

A. True

B. False

What is the rule for 30, 27, 24, 21?

- A.** Add 3
- B.** Multiply by 3
- C.** Subtract 3
- D.** Subtract 2

What comes next?

7, 14, 21, 28, ____

Next: _____

True or false?

The pattern 2, 4, 8, 16 adds 2 each time.

A. True

B. False

SET 12 • 4.OA.C.5

BACKWARD 2

Add or subtract 10 or 100 mentally • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8

What is 10 more than 347?

Answer: _____

True or false?

100 less than 612 is 512.

A. True

B. False

What is 100 more than 285?

- A.** 295
- B.** 385
- C.** 185
- D.** 286

What is 10 less than 503?

Answer: _____

True or false?

10 more than 790 is 890.

A. True

B. False

SET 12 • 4.OA.C.5

FORWARD 1

Form ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 1 and adds 1. Pattern B starts at 5 and adds 5.

Write the first four ordered pairs (A, B).

Ordered pairs: _____

True or false?

Pattern A is 0, 3, 6, 9 and Pattern B is 0, 9, 18, 27.
Each B term is 3 times the matching A term.

A. True

B. False

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8.

What is the B term when the A term is 6?

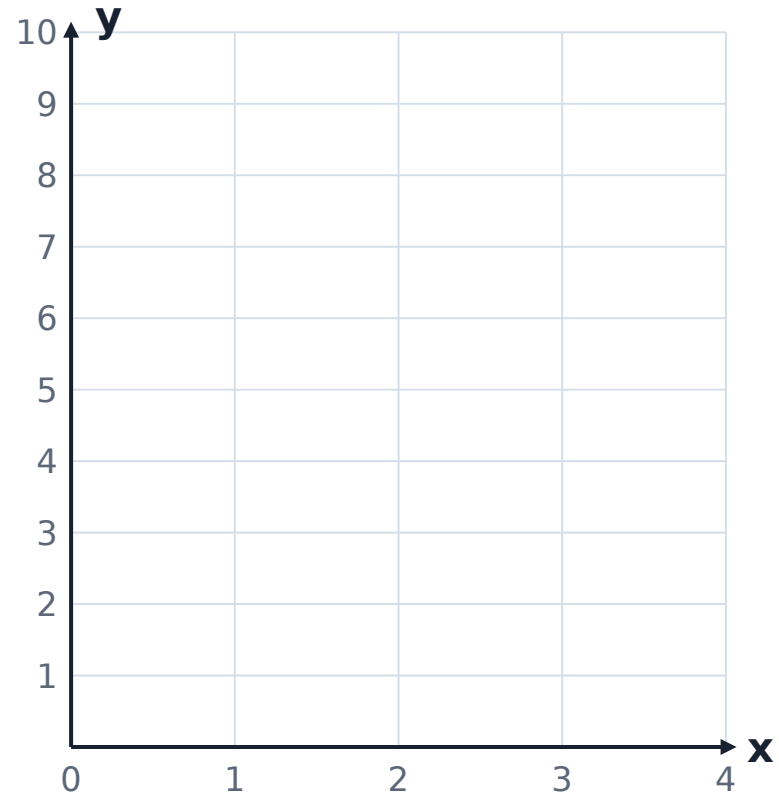
A. 14

B. 12

C. 48

D. 24

Graph the ordered pairs (0, 0), (1, 3), (2, 6), and (3, 9) from two patterns.



True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. The 4th pair is (20, 40).

- A. True
- B. False

SET 12 • 4.OA.C.5

FORWARD 2

Write an equation for a pattern with dependent and independent variables •

Grade 6 • Expressions & Equations • 6.EE.C.9

Figure 1 has 4 tiles, and each figure has 4 more tiles than the one before.

Write an equation for the tiles t in figure n .

Equation: _____

In $t = 4n$, which variable depends on the other?

- A.** t
- B.** n
- C.** 4
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	6	12	18	24

Equation: _____

True or false?

In $c = 8h$, when h increases by 1, c increases by 8.

- A. True
- B. False

True or false?

The table x : 1, 2, 3 and y : 3, 5, 7 fits $y = 3x$.

A. True

B. False

Identify features of a pattern that the rule does not state, and explain them

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Explain patterns in the multiplication and addition tables	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Explain the relationship between two patterns	5.OA.B.3
FORWARD 2	Evaluate expressions for a given value of the variable	6.EE.A.2.c

SET 13 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 4," starting at 2: 2, 6, 10, 14, 18.

What do you notice about every term? Explain why it happens.

Feature: _____

Why: _____

The rule is "add 5," starting at 3: 3, 8, 13, 18, 23.

Which statement is true?

- A.** Every term is a multiple of 5.
- B.** Every term is even.
- C.** The ones digits alternate between 3 and 8.
- D.** Every term is odd.

True or false?

The rule "add 2," starting at 7, makes only odd numbers.

A. True

B. False

True or false?

The rule "add 6," starting at 1, makes terms that alternate between odd and even.

- A. True
- B. False

The rule is "multiply by 2," starting at 3: 3, 6, 12, 24, 48.
After the first term, are the terms odd or even? Explain why.

Answer: _____

Why: _____

SET 13 • 4.OA.C.5

BACKWARD 1

Explain patterns in the multiplication and addition tables • Grade 3 •
Operations & Algebraic Thinking • 3.OA.D.9

True or false?

4 times any whole number is even.

A. True

B. False

Why is every product in the 6s row of the multiplication table even?

- A.** The 6s row has 10 products.
- B.** 6 is greater than 5.
- C.** Every number is even.
- D.** 6 is 2×3 , so every product is 2 times a number.

The 10s row is 10, 20, 30, 40, ...

What is the same about every product?

Answer: _____

True or false?

Every product in the 9s row of the multiplication table is odd.

- A.** True
- B.** False

Is the sum of two odd numbers odd or even?

Answer: _____

SET 13 • 4.OA.C.5

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

31 is odd.

A. True

B. False

Is 46 odd or even?

Answer: _____

Which number is odd?

- A.** 8
- B.** 14
- C.** 30
- D.** 27

Write 18 as the sum of two equal addends.

Sum: _____

True or false?

Adding two even numbers can give an odd number.

A. True

B. False

SET 13 • 4.OA.C.5

FORWARD 1

Explain the relationship between two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6. Pattern B starts at 0 and adds 2.

Explain why each term of A is 3 times the matching term of B.

Explanation: _____

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8. Each B term is 2 times the A term because B adds twice as much at every step.

A. True

B. False

Pattern A starts at 2 and adds 2. Pattern B starts at 2 and adds 6.

Why is B NOT always 3 times A?

- A.** A has more terms.
- B.** Both rules add even numbers.
- C.** B adds less than A.
- D.** They start at the same number, not at 0.

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

What is the B term when the A term is 35?

Term: _____

True or false?

Two patterns start at 0 and add 5 and 10. Each B term is 5 more than the matching A term.

A. True

B. False

SET 13 • 4.OA.C.5

FORWARD 2

Evaluate expressions for a given value of the variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $4n + 2$ when $n = 6$.

Value: _____

True or false?

The expression $2n$ gives an even number for every whole number n .

A. True

B. False

The n th term of 5, 9, 13, 17, ... is $4n + 1$.

What is the 10th term?

- A.** 40
- B.** 45
- C.** 41
- D.** 14

Evaluate $3n - 1$ when $n = 12$.

Value: _____

True or false?

When $n = 5$, $2n + 1 = 12$.

A. True

B. False

A digit in one place is worth ten times what it is worth in the place to its right

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	Hundreds, tens, and ones	2.NBT.A.1
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Place value in decimals: 10 times and 1/10 of a digit	5.NBT.A.1
FORWARD 2	Write and evaluate powers of 10 with exponents	6.EE.A.1

SET 14 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

True or false?

In 4,440, the 4 in the hundreds place is worth 10 times the 4 in the tens place.

- A. True
- B. False

The 8 in 8,325 is worth how many times the 8 in 825?

- A.** 10 times
- B.** 8 times
- C.** The same
- D.** 100 times

What is the value of the 6 in 62,904?

Value: _____

Write the number that is 10 times as much as 3,070.

Number: _____

True or false?

In 5,555, the 5 in the tens place is worth 10 times the 5 in the hundreds place.

- A. True
- B. False

SET 14 • 4.NBT.A.1

BACKWARD 1

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

How many tens make 1 hundred?

Tens: _____

True or false?

In 472, the 7 means 7 tens.

A. True

B. False

What is the value of the 3 in 360?

A. 3,000

B. 30

C. 3

D. 300

Write the number with 5 hundreds, 0 tens, and 8 ones.

Number: _____

True or false?

In 914, the 1 means 1 hundred.

A. True

B. False

SET 14 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

Multiply.

$$6 \times 40$$

Product: _____

True or false?

$$3 \times 70 = 210$$

A. True

B. False

$$9 \times 50 = ?$$

A. 45

B. 59

C. 4,500

D. 450

Multiply.

$$8 \times 30$$

Product: _____

True or false?

$$5 \times 80 = 4,000$$

A. True

B. False

SET 14 • 4.NBT.A.1

FORWARD 1

Place value in decimals: 10 times and $\frac{1}{10}$ of a digit • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1

True or false?

In 3.33, the 3 in the ones place is worth 10 times the 3 in the tenths place.

- A. True
- B. False

What is $\frac{1}{10}$ of 70?

Answer: _____

In 6.66, the 6 in the hundredths place is worth what part of the 6 in the tenths place?

A. $\frac{1}{10}$

B. The same amount

C. 10 times as much

D. $\frac{1}{100}$

What is 10 times 0.4?

Answer: _____

True or false?

In 2.22, the 2 in the tenths place is worth 10 times the 2 in the ones place.

A. True

B. False

SET 14 • 4.NBT.A.1

FORWARD 2

Write and evaluate powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $10 \times 10 \times 10$ using an exponent.

Exponent form: _____

True or false?

$$10^4 = 10,000$$

A. True

B. False

Evaluate.

$$10^2$$

- A.** 12
- B.** 20
- C.** 1,000
- D.** 100

Evaluate.

$$4 \times 10^3$$

Value: _____

True or false?

10^5 has 4 zeros.

A. True

B. False

Use place value and division to explain why one number is ten times another

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	A hundred is a bundle of ten tens	2.NBT.A.1.a
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Divide whole numbers and decimals by powers of 10	5.NBT.A.2
FORWARD 2	Divide multi-digit decimals	6.NS.B.3

SET 15 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Use place value to find $5,000 \div 500$.

Quotient: _____

True or false?

$900 \div 90 = 10$ because 9 hundreds is 10 times 9 tens.

A. True

B. False

Which statement explains why $3,000 \div 300 = 10$?

- A.** $3,000 - 300 = 10$
- B.** 3 thousands is 10 times 3 hundreds.
- C.** 3,000 has one more zero.
- D.** $300 \div 3,000 = 10$

4,000 is how many times as much as 400? Explain using place value.

Answer: _____

Explanation: _____

True or false?

$$60 \div 6 = 100$$

A. True

B. False

SET 15 • 4.NBT.A.1

BACKWARD 1

A hundred is a bundle of ten tens • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a

True or false?

100 can be thought of as a bundle of 10 tens.

A. True

B. False

How many tens are in 300?

Tens: _____

How many hundreds are in 800?

- A.** 8
- B.** 800
- C.** 18
- D.** 80

How many tens are in 50?

Tens: _____

True or false?

400 is 4 tens.

A. True

B. False

SET 15 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$4 \times 90$$

Product: _____

True or false?

$$7 \times 60 = 420$$

A. True

B. False

$$2 \times 80 = ?$$

- A.** 160
- B.** 82
- C.** 16
- D.** 1,600

Multiply.

$$9 \times 20$$

Product: _____

True or false?

$$6 \times 50 = 3,000$$

A. True

B. False

SET 15 • 4.NBT.A.1

FORWARD 1

Divide whole numbers and decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$$4,500 \div 100$$

Quotient: _____

True or false?

$$7,000 \div 10^3 = 7$$

A. True

B. False

Divide.

$$36 \div 10$$

A. 360

B. 3.6

C. 36

D. 0.36

Divide.

$$820 \div 10^2$$

Quotient: _____

True or false?

$$50 \div 10 = 500$$

A. True

B. False

SET 15 • 4.NBT.A.1

FORWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.5 \div 0.45$$

Quotient: _____

True or false?

$$0.8 \div 0.08 = 10$$

A. True

B. False

Divide.

$$6 \div 0.6$$

A. 10

B. 0.1

C. 1

D. 100

Divide.

$$2.4 \div 0.3$$

Quotient: _____

True or false?

$$1.2 \div 0.12 = 100$$

A. True

B. False

Read and write multi-digit whole numbers using numerals and number names

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Read and write numbers to 1000	2.NBT.A.3
BACKWARD 2	Hundreds, tens, and ones	2.NBT.A.1
FORWARD 1	Read and write decimals to thousandths	5.NBT.A.3.a
FORWARD 2	Use positive and negative numbers to describe quantities — near...	6.NS.C.5

SET 16 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write the number in standard form.

forty-six thousand, two hundred eight

Number: _____

Write 309,051 in words.

Words: _____

Which is 72,016 in words?

- A.** seventy-two thousand, sixteen
- B.** seven thousand, two hundred sixteen
- C.** seventy-two thousand, one hundred six
- D.** seventy-two thousand, one hundred sixty

True or false?

Six hundred thousand, forty is written 600,400.

- A.** True
- B.** False

Write the number in standard form.

ninety thousand, nine hundred

Number: _____

SET 16 • 4.NBT.A.2

BACKWARD 1

Read and write numbers to 1000 • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.3

Write the number in standard form.

four hundred seventeen

Number: _____

True or false?

Six hundred five is written 650.

A. True

B. False

Which is 238 in words?

- A.** two hundred thirty-eight
- B.** three hundred twenty-eight
- C.** two hundred eighty-three
- D.** twenty-three eight

Write 702 in words.

Words: _____

True or false?

Nine hundred ninety is written 990.

A. True

B. False

SET 16 • 4.NBT.A.2

BACKWARD 2

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

What digit is in the tens place of 586?

Digit: _____

True or false?

In 249, the 2 is worth 200.

A. True

B. False

Which number has 7 hundreds, 3 tens, and 0 ones?

A. 730

B. 703

C. 73

D. 370

How many hundreds are in 600?

Hundreds: _____

True or false?

In 381, the 8 is worth 8.

A. True

B. False

SET 16 • 4.NBT.A.2

FORWARD 1

Read and write decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a

Write the number in standard form.

four and twenty-five thousandths

Number: _____

True or false?

0.6 is read "six tenths."

A. True

B. False

Which is 3.48 in words?

- A.** thirty-four and eight tenths
- B.** three hundred forty-eight
- C.** three and forty-eight tenths
- D.** three and forty-eight hundredths

Write 12.307 in words.

Words: _____

True or false?

Five and nine hundredths is written 5.9.

A. True

B. False

FORWARD 2

Use positive and negative numbers to describe quantities • Grade 6 • The Number System • 6.NS.C.5

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A diver is 30 feet below sea level.

Write the diver's position as an integer.

Integer: _____

True or false?

A gain of 8 yards can be written as +8.

A. True

B. False

Which integer shows a temperature of 12 degrees below zero?

A. -12

B. 12

C. 0

D. -21

A deposit of \$45 is written as +45.

How is a withdrawal of \$45 written?

Integer: _____

True or false?

–5 means 5 more than zero.

A. True

B. False

Write multi-digit whole numbers in expanded form

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Expanded form for numbers to 1000	2.NBT.A.3
BACKWARD 2	Two-digit numbers as tens and ones	1.NBT.B.2
FORWARD 1	Write decimals in expanded form	5.NBT.A.3.a
FORWARD 2	Write numbers using powers of 10 with exponents	6.EE.A.1

SET 17 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write 7,345 in expanded form.

Expanded form: _____

Write the number.

$$50,000 + 2,000 + 80 + 6$$

Number: _____

Which is the expanded form of 40,209?

A. $40,000 + 2,000 + 9$

B. $40,000 + 20 + 9$

C. $40,000 + 200 + 9$

D. $4,000 + 200 + 9$

True or false?

$$600,000 + 3,000 + 70 = 630,070$$

A. True

B. False

Write 18,504 in expanded form.

Expanded form: _____

SET 17 • 4.NBT.A.2

BACKWARD 1

Expanded form for numbers to 1000 • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3

Write 846 in expanded form.

Expanded form: _____

True or false?

$$300 + 9 = 390$$

A. True

B. False

Which number is $500 + 20 + 7$?

- A.** 527
- B.** 572
- C.** 257
- D.** 5,207

Write the number.

$$900 + 60$$

Number: _____

True or false?

$$200 + 40 + 1 = 241$$

A. True

B. False

SET 17 • 4.NBT.A.2

BACKWARD 2

Two-digit numbers as tens and ones • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2

How many tens and ones are in 47?

Answer: _____

True or false?

63 is 6 tens and 3 ones.

A. True

B. False

Which number is 8 tens and 2 ones?

- A.** 28
- B.** 10
- C.** 82
- D.** 802

Write the number that is 5 tens and 0 ones.

Number: _____

True or false?

91 is 1 ten and 9 ones.

A. True

B. False

FORWARD 1

Write decimals in expanded form • Grade 5 • Number & Operations in Base Ten
• 5.NBT.A.3.a

Write 3.46 in expanded form.

Expanded form: _____

True or false?

$$2 + 0.5 + 0.03 = 2.53$$

A. True

B. False

Which number is $7 \times 1 + 2 \times \frac{1}{10} + 9 \times \frac{1}{1000}$?

A. 7.029

B. 7.209

C. 72.9

D. 7.29

Write the number.

$$5 \times 10 + 1 \times 1 + 8 \times \frac{1}{100}$$

Number: _____

True or false?

$$4 \times \frac{1}{10} + 1 \times \frac{1}{100} = 0.041$$

A. True

B. False

SET 17 • 4.NBT.A.2

FORWARD 2

Write numbers using powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $6,000 + 400 + 70 + 2$ using powers of 10.

Expression: _____

True or false?

$$3 \times 10^4 = 30,000$$

A. True

B. False

Which number equals $5 \times 10^3 + 8 \times 10$?

- A.** 580
- B.** 5,008
- C.** 5,800
- D.** 5,080

Evaluate.

$$2 \times 10^3 + 9 \times 10^2$$

Value: _____

True or false?

$$7 \times 10^2 = 70$$

A. True

B. False

Compare multi-digit whole numbers using $>$, $=$, and $<$

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Compare three-digit numbers	2.NBT.A.4
BACKWARD 2	Compare two-digit numbers	1.NBT.B.3
FORWARD 1	Compare decimals to thousandths	5.NBT.A.3.b
FORWARD 2	Order rational numbers using their positions on a number line	6.NS.C.7.a

SET 18 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write $>$, $=$, or $<$.

45,621 ____ 45,612

Symbol: _____

Which comparison is true?

A. $80,099 > 80,100$

B. $80,099 = 80,100$

C. $80,100 < 80,099$

D. $80,099 < 80,100$

True or false?

305,000 < 350,000

A. True

B. False

Order from least to greatest.

27,450 24,750 27,045

Order: _____

True or false?

9,998 > 10,001

A. True

B. False

SET 18 • 4.NBT.A.2

BACKWARD 1

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

482 428

Symbol: _____

True or false?

$$309 < 390$$

A. True

B. False

Which number is greatest?

A. 170

B. 699

C. 701

D. 710

Write $>$, $=$, or $<$.

555 555

Symbol: _____

True or false?

$$640 > 664$$

A. True

B. False

SET 18 • 4.NBT.A.2

BACKWARD 2

Compare two-digit numbers • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Write $>$, $=$, or $<$.

67 76

Symbol: _____

True or false?

$$58 > 49$$

A. True

B. False

Which number is less than 35?

A. 35

B. 53

C. 29

D. 40

Write $>$, $=$, or $<$.

90 19

Symbol: _____

True or false?

$$42 < 24$$

A. True

B. False

SET 18 • 4.NBT.A.2

FORWARD 1

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Write $>$, $=$, or $<$.

0.45 0.405

Symbol: _____

True or false?

$$3.09 < 3.1$$

A. True

B. False

Which decimal is greatest?

- A.** 2.5
- B.** 2.09
- C.** 2.405
- D.** 2.48

Write $>$, $=$, or $<$.

6.70 6.7

Symbol: _____

True or false?

$$0.125 > 0.13$$

A. True

B. False

SET 18 • 4.NBT.A.2

FORWARD 2

Order rational numbers using their positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

Write $>$, $=$, or $<$.

$$-8 \text{ ____ } -3$$

Symbol: _____

True or false?

$$-2 > -10$$

A. True

B. False

Which number is least?

A. 0

B. -5

C. 5

D. -15

Order from least to greatest.

4, -6, 0, -1

Order: _____

True or false?

$$-7 > 2$$

A. True

B. False

Round multi-digit whole numbers to any place

MAIN	Grade 4 main question	4.NBT.A.3
BACKWARD 1	Round to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Skip-count by 5s, 10s, and 100s	2.NBT.A.2
FORWARD 1	Round decimals to any place	5.NBT.A.4
FORWARD 2	Compute with decimals and round the result — nearest related (n...	6.NS.B.3

SET 19 • 4.NBT.A.3

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 46,829 to the nearest thousand.

Rounded: _____

Round 352,618 to the nearest ten thousand.

Rounded: _____

Round 7,950 to the nearest hundred.

- A.** 7,900
- B.** 7,960
- C.** 7,000
- D.** 8,000

True or false?

128,476 rounded to the nearest hundred thousand is 200,000.

- A. True
- B. False

A stadium has 63,548 seats.

Round the number of seats to the nearest thousand.

Rounded: _____

SET 19 • 4.NBT.A.3

BACKWARD 1

Round to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.1

Round 84 to the nearest ten.

Rounded: _____

True or false?

652 rounded to the nearest hundred is 700.

A. True

B. False

Round 345 to the nearest ten.

A. 400

B. 350

C. 300

D. 340

Round 961 to the nearest hundred.

Rounded: _____

True or false?

26 rounded to the nearest ten is 20.

A. True

B. False

SET 19 • 4.NBT.A.3

BACKWARD 2

Skip-count by 5s, 10s, and 100s • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.2

Skip-count by 100s.

300, 400, 500, ____

Next: _____

True or false?

Counting by 10s from 470 gives 480, 490, 500.

A. True

B. False

What comes next when counting by 5s?

85, 90, 95, ____

A. 96

B. 105

C. 100

D. 99

Skip-count by 10s.

620, 630, 640, ____

Next: _____

True or false?

Counting by 100s from 700 gives 800, 900, 990.

A. True

B. False

SET 19 • 4.NBT.A.3

FORWARD 1

Round decimals to any place • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.468 to the nearest tenth.

Rounded: _____

Round 14.729 to the nearest whole number.

Rounded: _____

Round 0.638 to the nearest hundredth.

- A.** 0.6
- B.** 0.64
- C.** 0.7
- D.** 0.63

True or false?

5.95 rounded to the nearest tenth is 6.0.

A. True

B. False

True or false?

8.249 rounded to the nearest tenth is 8.3.

A. True

B. False

FORWARD 2

Compute with decimals and round the result • Grade 6 • The Number System •
6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Divide. Round the quotient to the nearest tenth.

$$10 \div 3$$

Quotient: _____

True or false?

$$7.25 \times 4 = 29$$

A. True

B. False

Divide. Round the quotient to the nearest hundredth.

$$2 \div 7$$

A. 0.27

B. 0.29

C. 0.28

D. 0.3

Multiply. Round the product to the nearest whole number.

$$3.6 \times 2.4$$

Product: _____

True or false?

$15 \div 4$, rounded to the nearest whole number, is 3.

- A. True
- B. False

Add multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Add within 1000	3.NBT.A.2
BACKWARD 2	Add by place value, making a ten or a hundred	2.NBT.B.7
FORWARD 1	Add decimals to hundredths	5.NBT.B.7
FORWARD 2	Add multi-digit decimals with the standard algorithm	6.NS.B.3

SET 20 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to add. Show your work.

$$4,587 + 2,946$$

Sum: _____

Use the standard algorithm to add. Show your work.

$$38,209 + 15,894$$

Sum: _____

Add.

$$6,078 + 3,945$$

- A.** 9,023
- B.** 9,923
- C.** 10,013
- D.** 10,023

A library had 12,486 books. It got 3,758 more.

Use the standard algorithm to find how many books it has now. Show your work.

Books: _____

True or false?

In the standard algorithm for $3,685 + 1,476$, you regroup 11 ones as 1 ten and 1 one.

- A. True
- B. False

SET 20 • 4.NBT.B.4

BACKWARD 1

Add within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Add.

$$356 + 478$$

Sum: _____

True or false?

$$609 + 295 = 904$$

A. True

B. False

$$147 + 268 = ?$$

- A.** 415
- B.** 315
- C.** 405
- D.** 425

Add.

$$523 + 389$$

Sum: _____

True or false?

$$480 + 360 = 740$$

A. True

B. False

SET 20 • 4.NBT.B.4

BACKWARD 2

Add by place value, making a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

7 ones and 5 ones make 1 ten and 2 ones.

A. True

B. False

Rename 16 tens as hundreds and tens.

Answer: _____

Add the hundreds, the tens, and the ones.

$$248 + 135$$

A. 483

B. 373

C. 383

D. 313

Add.

$$200 + 70 + 4 + 300 + 20 + 9$$

Sum: _____

True or false?

12 ones is the same as 2 tens.

A. True

B. False

SET 20 • 4.NBT.B.4

FORWARD 1

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$35.68 + 7.45$$

Sum: _____

Add.

$$4.9 + 12.36$$

Sum: _____

Add.

$$0.75 + 0.48$$

- A.** 1.13
- B.** 12.3
- C.** 1.23
- D.** 0.123

True or false?

$$6.07 + 3.9 = 9.97$$

A. True

B. False

True or false?

$$2.5 + 1.75 = 3.80$$

A. True

B. False

SET 20 • 4.NBT.B.4

FORWARD 2

Add multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.406 + 75.89$$

Sum: _____

Add.

$$9.075 + 0.98$$

Sum: _____

Add.

$$46.3 + 8.759$$

A. 55.059

B. 131.89

C. 55.159

D. 54.059

True or false?

$$0.625 + 0.375 = 1$$

A. True

B. False

True or false?

$$12.48 + 3.6 = 12.84$$

A. True

B. False

Subtract multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Subtract within 1000	3.NBT.A.2
BACKWARD 2	Subtract by breaking apart a ten or a hundred	2.NBT.B.7
FORWARD 1	Subtract decimals to hundredths	5.NBT.B.7
FORWARD 2	Subtract multi-digit decimals with the standard algorithm	6.NS.B.3

SET 21 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to subtract. Show your work.

$$8,253 - 4,678$$

Difference: _____

Use the standard algorithm to subtract. Show your work.

$$60,000 - 27,438$$

Difference: _____

Subtract.

$$7,012 - 3,456$$

A. 4,656

B. 3,556

C. 4,444

D. 3,566

A school raised \$15,250. It spent \$8,795.

Use the standard algorithm to find how much money is left. Show your work.

Money left: _____

True or false?

To find $5,042 - 1,317$ with the standard algorithm, you rename 1 ten as 10 ones.

- A. True
- B. False

SET 21 • 4.NBT.B.4

BACKWARD 1

Subtract within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Subtract.

$$732 - 485$$

Difference: _____

True or false?

$$600 - 278 = 322$$

A. True

B. False

$$514 - 296 = ?$$

A. 222

B. 228

C. 218

D. 318

Subtract.

$$905 - 467$$

Difference: _____

True or false?

$$840 - 390 = 550$$

A. True

B. False

SET 21 • 4.NBT.B.4

BACKWARD 2

Subtract by breaking apart a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

To find $52 - 8$, you can rename 1 ten as 10 ones.

- A. True
- B. False

Rename 4 hundreds 2 tens as 3 hundreds and how many tens?

Tens: _____

Which shows 300 renamed so you can subtract 126?

- A.** 2 hundreds 10 ones
- B.** 3 hundreds 10 tens
- C.** 30 tens 1 one
- D.** 2 hundreds 9 tens 10 ones

Subtract.

$$400 - 158$$

Difference: _____

True or false?

6 tens 3 ones = 5 tens 3 ones

A. True

B. False

SET 21 • 4.NBT.B.4

FORWARD 1

Subtract decimals to hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Subtract.

$$42.5 - 18.76$$

Difference: _____

Subtract.

$$9 - 3.45$$

Difference: _____

Subtract.

$$6.03 - 2.8$$

A. 3.22

B. 3.75

C. 4.23

D. 3.23

True or false?

$$10 - 0.25 = 9.75$$

A. True

B. False

True or false?

$$5.4 - 2.15 = 3.35$$

A. True

B. False

SET 21 • 4.NBT.B.4

FORWARD 2

Subtract multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$$100 - 36.875$$

Difference: _____

Subtract.

$$15.06 - 8.3$$

Difference: _____

Subtract.

$$4.2 - 1.375$$

A. 2.825

B. 2.925

C. 3.825

D. 3.175

True or false?

$$50.5 - 0.505 = 49.995$$

A. True

B. False

True or false?

$$8 - 0.08 = 0.72$$

A. True

B. False

Multiply a whole number of up to four digits by a one-digit number

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Multiplication facts within 100	3.OA.C.7
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Multiply multi-digit whole numbers using the standard algorithm	5.NBT.B.5
FORWARD 2	Multiply multi-digit decimals	6.NS.B.3

SET 22 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$4,236 \times 7$$

Product: _____

Multiply.

$$608 \times 5$$

Product: _____

Multiply.

$$3,049 \times 6$$

- A.** 3,055
- B.** 18,254
- C.** 18,924
- D.** 18,294

A theater sells 1,275 tickets each night for 8 nights.

How many tickets does it sell?

Tickets: _____

True or false?

$$2,513 \times 4 = 10,042$$

A. True

B. False

SET 22 • 4.NBT.B.5

BACKWARD 1

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

True or false?

$$9 \times 8 = 72$$

A. True

B. False

$$4 \times 7 = ?$$

A. 24

B. 32

C. 11

D. 28

Multiply.

$$6 \times 9$$

Product: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

SET 22 • 4.NBT.B.5

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$7 \times 80$$

Product: _____

True or false?

$$5 \times 60 = 300$$

A. True

B. False

$$3 \times 90 = ?$$

A. 27

B. 2,700

C. 93

D. 270

Multiply.

$$4 \times 70$$

Product: _____

True or false?

$$8 \times 40 = 3,200$$

A. True

B. False

SET 22 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$3,214 \times 26$$

Product: _____

Multiply.

$$507 \times 48$$

Product: _____

Multiply.

$$1,250 \times 34$$

A. 42,500

B. 8,750

C. 42,050

D. 41,500

Multiply.

$$289 \times 75$$

Product: _____

True or false?

$$406 \times 52 = 21,112$$

A. True

B. False

SET 22 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$4.236 \times 7$$

Product: _____

Multiply.

$$6.08 \times 0.5$$

Product: _____

Multiply.

$$30.49 \times 0.6$$

- A.** 18.94
- B.** 18.294
- C.** 182.94
- D.** 1.8294

True or false?

$$2.5 \times 1.6 = 4$$

A. True

B. False

True or false?

$$0.7 \times 0.08 = 0.56$$

A. True

B. False

Multiply two two-digit numbers

MAIN

Grade 4 main question

4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10

3.NBT.A.3

BACKWARD 2

Break apart a factor to multiply

3.OA.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm

5.NBT.B.5

FORWARD 2

Multiply multi-digit decimals

6.NS.B.3

SET 23 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 27$$

Product: _____

Multiply.

$$58 \times 63$$

Product: _____

Multiply.

$$46 \times 19$$

A. 414

B. 884

C. 864

D. 874

A box holds 24 cans.

How many cans are in 36 boxes?

Cans: _____

True or false?

$$72 \times 15 = 1,008$$

A. True

B. False

SET 23 • 4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

Multiply.

$$3 \times 40$$

Product: _____

True or false?

$$6 \times 90 = 540$$

A. True

B. False

$$8 \times 20 = ?$$

- A.** 28
- B.** 1,600
- C.** 160
- D.** 16

Multiply.

$$9 \times 70$$

Product: _____

True or false?

$$4 \times 50 = 2,000$$

A. True

B. False

SET 23 • 4.NBT.B.5

BACKWARD 2

Break apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

True or false?

$$8 \times 7 = 8 \times 5 + 8 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 9 = 6 \times 5 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 4×8 ?

A. $4 \times 5 \times 3$

B. $4 \times 5 + 4 \times 3$

C. $4 \times 5 + 3$

D. $4 + 5 \times 3$

Find 7×6 by adding $7 \times 3 + 7 \times 3$.

Product: _____

True or false?

$$9 \times 6 = 9 \times 3 + 3$$

A. True

B. False

SET 23 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$427 \times 53$$

Product: _____

Multiply.

$$86 \times 145$$

Product: _____

Multiply.

$$312 \times 64$$

A. 18,968

B. 19,968

C. 19,868

D. 1,248

Multiply.

$$2,005 \times 18$$

Product: _____

True or false?

$$95 \times 95 = 9,025$$

A. True

B. False

SET 23 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.4 \times 2.7$$

Product: _____

Multiply.

$$0.58 \times 6.3$$

Product: _____

Multiply.

$$4.6 \times 0.19$$

- A.** 4.79
- B.** 0.874
- C.** 0.0874
- D.** 8.74

True or false?

$$2.4 \times 3.6 = 8.64$$

A. True

B. False

True or false?

$$7.2 \times 1.5 = 1.08$$

A. True

B. False

Illustrate and explain multiplication with equations, arrays, and area models

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Use area to show the distributive property	3.MD.C.7.c
BACKWARD 2	Break apart an array to multiply	3.OA.B.5
FORWARD 1	Use place value and area models to multiply decimals	5.NBT.B.7
FORWARD 2	Use the distributive property to write a sum as a product	6.NS.B.4

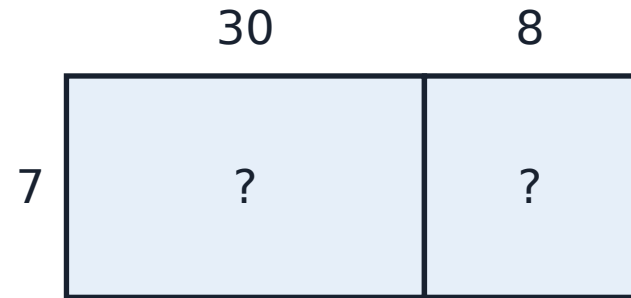
SET 24 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Complete the area model for 7×38 by writing the area of each part.

Then write the product.



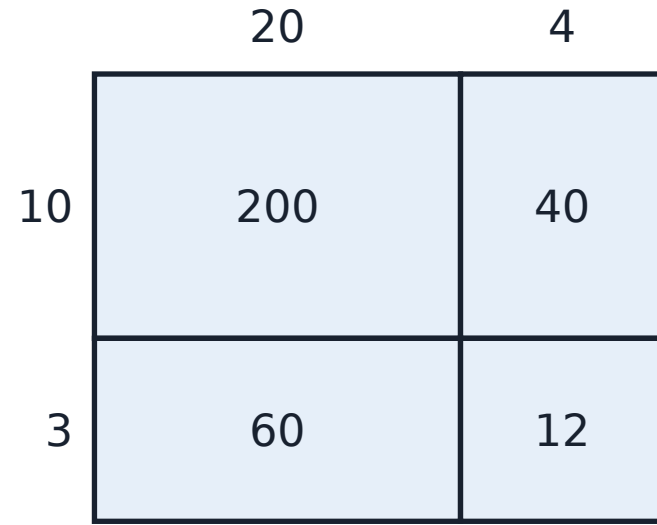
Product: _____

The area model shows
 24×13 .

Write the four partial
products and the total.

Partial products: _____

Total: _____



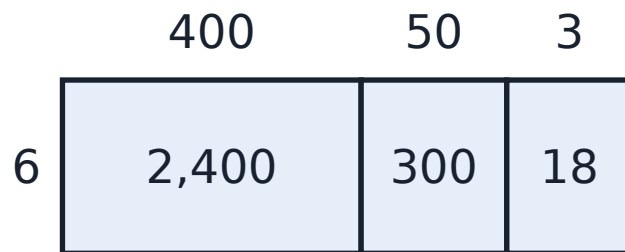
Which equation matches the area model?

A. $6 \times 453 = 2,400 + 50 + 3$

B. $6 \times 453 = 2,400 + 300 + 18$

C. $6 \times 453 = 6 \times 400 + 53$

D. $6 + 453 = 2,718$



Use an area model or equations to multiply. Show your model or equations.

$$9 \times 2,146$$

Product: _____

Explain why $15 \times 12 = 15 \times 10 + 15 \times 2$.

Explanation: _____

SET 24 • 4.NBT.B.5

BACKWARD 1

Use area to show the distributive property • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

A rectangle that is 4 by $(5 + 3)$ has an area of $4 \times 5 + 4 \times 3$.

A. True

B. False

A 6-by-9 rectangle is split into a 6-by-5 part and a 6-by-4 part.

What is the total area?

Area: _____

A 3-by-8 rectangle is split into a 3-by-5 part and a 3-by-3 part.

Which expression gives its area?

A. $3 \times 8 + 3$

B. $3 \times 5 \times 3$

C. $3 + 5 + 3$

D. $3 \times 5 + 3 \times 3$

Fill in the blank.

$$7 \times 9 = 7 \times 5 + 7 \times \underline{\quad}$$

Blank: _____

True or false?

$$5 \times 7 = 5 \times 4 + 3$$

A. True

B. False

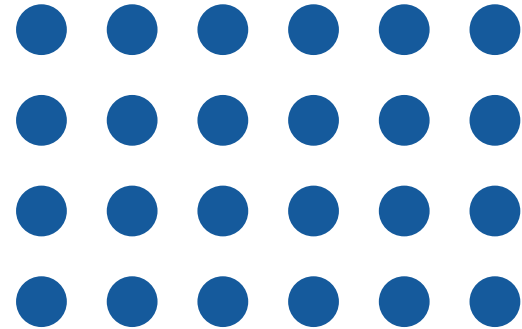
SET 24 • 4.NBT.B.5

BACKWARD 2

Break apart an array to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

Write a multiplication equation for the array.

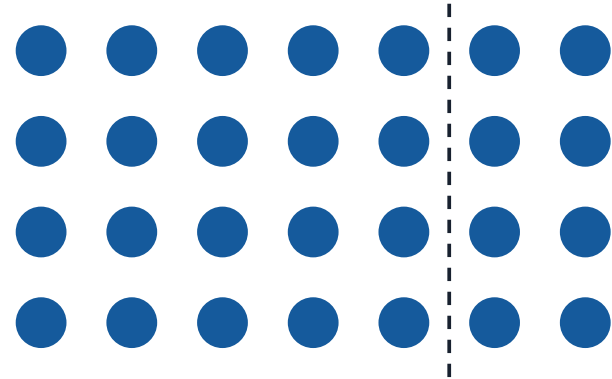
Equation: _____



The dashed line splits the array into two parts.

Which equation matches?

- A.** $4 \times 7 = 4 + 5 + 2$
- B.** $4 \times 7 = 4 \times 5 \times 2$
- C.** $4 \times 7 = 4 \times 5 + 4 \times 2$
- D.** $4 + 7 = 4 \times 5 + 2$



True or false?

An array of 3 rows of 9 can be split into 3 rows of 5 and 3 rows of 4.

A. True

B. False

An array of 6 rows of 8 is split into 6 rows of 5 and 6 rows of 3.

How many dots are in the whole array?

Dots: _____

True or false?

An array of 5 rows of 6, split into two arrays of 5 rows of 3, has 33 dots.

A. True

B. False

SET 24 • 4.NBT.B.5

FORWARD 1

Use place value and area models to multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

An area model splits 3×2.4 into 3×2 and 3×0.4 .

What is the product?

Product: _____

True or false?

$$4 \times 1.25 = 4 \times 1 + 4 \times 0.25 = 5$$

A. True

B. False

Which shows 6×3.5 broken apart by place value?

A. $6 \times 3 \times 0.5$

B. $6 \times 3 + 0.5$

C. $6 + 3 + 0.5$

D. $6 \times 3 + 6 \times 0.5$

Multiply.

$$5 \times 0.36$$

Product: _____

True or false?

$$2 \times 4.3 = 8.3$$

A. True

B. False

SET 24 • 4.NBT.B.5

FORWARD 2

Use the distributive property to write a sum as a product • Grade 6 • The Number System • 6.NS.B.4

Use the factor 4 to write $36 + 8$ as a product.

Expression: _____

True or false?

$$7(10 + 3) = 70 + 21$$

A. True

B. False

Which expression equals 5×47 ?

A. $5(40 + 7)$

B. $5 \times 40 + 7$

C. $5 + 40 \times 7$

D. $(5 + 40) \times 7$

Use the greatest common factor to write $24 + 60$ as a product.

Expression: _____

True or false?

$$3(12 + 5) = 36 + 5$$

A. True

B. False

Divide up to four-digit dividends by one-digit divisors, with remainders

MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division facts within 100	3.OA.C.7
BACKWARD 2	Share equally	3.OA.A.2
FORWARD 1	Divide with two-digit divisors	5.NBT.B.6
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 25 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$952 \div 7$$

Quotient: _____

Divide.

$$4,815 \div 5$$

Quotient: _____

Divide.

$$2,184 \div 6$$

- A.** 364
- B.** 354
- C.** 2,178
- D.** 374

Divide. Write the quotient and the remainder.

$$1,357 \div 4$$

Quotient: _____

Remainder: _____

True or false?

$$3,600 \div 9 = 40$$

A. True

B. False

SET 25 • 4.NBT.B.6

BACKWARD 1

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

True or false?

$$72 \div 8 = 9$$

A. True

B. False

$$54 \div 6 = ?$$

A. 48

B. 7

C. 8

D. 9

Divide.

$$27 \div 3$$

Quotient: _____

True or false?

$$63 \div 9 = 8$$

A. True

B. False

SET 25 • 4.NBT.B.6

BACKWARD 2

Share equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

28 marbles are shared equally by 4 children.

How many marbles does each child get?

Marbles: _____

True or false?

When 45 is split into 5 equal groups, each group has 9.

A. True

B. False

Which expression shows 32 shared equally among 8 people?

A. $32 - 8$

B. 32×8

C. $32 \div 8$

D. $8 \div 32$

48 stickers are put into groups of 6.

How many groups are there?

Groups: _____

True or false?

20 shared equally by 4 people gives each person 16.

A. True

B. False

SET 25 • 4.NBT.B.6

FORWARD 1

Divide with two-digit divisors • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,736 \div 28$$

Quotient: _____

Divide.

$$5,904 \div 72$$

Quotient: _____

Divide.

$$3,360 \div 48$$

A. 70

B. 7

C. 68

D. 700

Divide. Write the quotient and the remainder.

$$2,000 \div 37$$

Quotient: _____

Remainder: _____

True or false?

$$1,125 \div 25 = 54$$

A. True

B. False

SET 25 • 4.NBT.B.6

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$9,516 \div 39$$

Quotient: _____

Divide.

$$27,648 \div 72$$

Quotient: _____

Divide.

$$15,525 \div 45$$

- A.** 355
- B.** 345
- C.** 3,450
- D.** 335

Divide.

$$8,184 \div 62$$

Quotient: _____

True or false?

$$20,808 \div 51 = 408$$

A. True

B. False

Illustrate and explain division with equations, area models, and place value

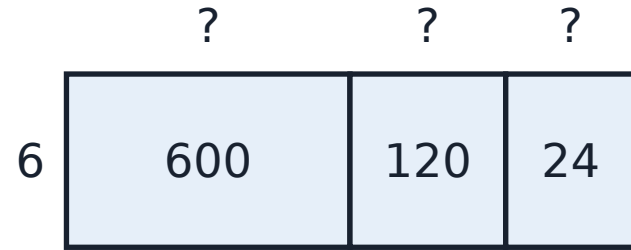
MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find a missing side from the area of a rectangle	3.MD.C.7.b
FORWARD 1	Explain division with two-digit divisors using area models and equ...	5.NBT.B.6
FORWARD 2	Connect the standard division algorithm to partial quotients	6.NS.B.2

SET 26 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Complete the area model for $744 \div 6$ by writing the missing length of each part. Then write the quotient.



Quotient: _____

Explain how these equations show $852 \div 4$.

$$4 \times 200 = 800$$

$$4 \times 13 = 52$$

$$800 + 52 = 852$$

Quotient: _____

Explanation: _____

Ana finds $3,515 \div 5$ by thinking $3,500 \div 5 = 700$ and $15 \div 5 = 3$.

What is the quotient?

A. 7,003

B. 73

C. 703

D. 730

Use an area model or equations to divide. Show your model or equations.

$$1,792 \div 8$$

Quotient: _____

True or false?

To find $636 \div 3$, you can divide each place:
 $600 \div 3 = 200$, $30 \div 3 = 10$, and $6 \div 3 = 2$. The quotient is 212.

- A.** True
- B.** False

SET 26 • 4.NBT.B.6

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $56 \div 8$ by thinking $8 \times ? = 56$.

Quotient: _____

True or false?

$36 \div 9 = 4$ because $9 \times 4 = 36$.

A. True

B. False

Which fact helps you find $45 \div 5$?

A. $5 \times 9 = 45$

B. $45 - 5 = 40$

C. $9 + 5 = 14$

D. $5 \times 5 = 25$

Find the unknown factor.

$$7 \times ? = 63$$

Factor: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 26 • 4.NBT.B.6

BACKWARD 2

Find a missing side from the area of a rectangle • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle has an area of 48 square units. One side is 6 units long.

How long is the other side?

Length: _____

True or false?

A rectangle that is 7 units by 5 units has an area of 35 square units.

- A.** True
- B.** False

A rectangle has an area of 36 square feet and a width of 4 feet.

What is its length?

A. 32 feet

B. 40 feet

C. 9 feet

D. 8 feet

What is the area of a rectangle that is 8 cm by 6 cm?

Area: _____

True or false?

A rectangle with an area of 30 square units and one side of 5 units has another side of 25 units.

- A. True
- B. False

SET 26 • 4.NBT.B.6

FORWARD 1

Explain division with two-digit divisors using area models and equations •
Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6